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1. Jika kapasitansi sebuah kapasitor berisi udara 7,4 pF. Bahan dielektrik dengan K berapa yang harus disisipkan ke dalam kapasitor sehingga dapat menyimpan energi sebesar 7,4 μJ dengan beda potensial maksimum 652 V?

Dik : $C_0 = 7{,}4\ pF = 7{,}4 \cdot 10^{-12}\ F$
$U = 7{,}4\ \mu J = 7{,}4 \cdot 10^{-6}\ J$
$V_{maks} = 652\ V$
Dit : K?
Jawab : $U = \frac{1}{2} CV^2 = \frac{1}{2} k \cdot C_0 \cdot V^2$
maka : $K = \frac{2U}{C_0 \cdot V^2}$

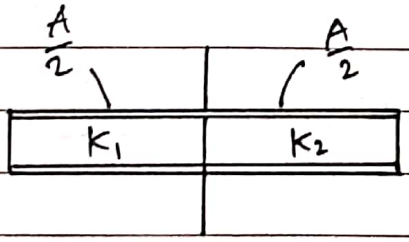
$$K = \frac{2(7{,}4 \cdot 10^{-6})}{(7{,}4 \cdot 10^{-12})(652)^2}$$

$$K = \frac{14{,}8 \cdot 10^{-6}}{(7{,}4 \cdot 10^{-12})(425.104)}$$

$$K = \frac{14{,}8 \cdot 10^{-6}}{3{,}146 \cdot 10^{-6}}$$

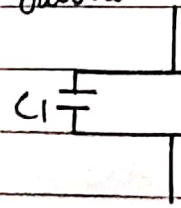
$$K = 4{,}705 \ //$$

2.



Dik : $A = 5{,}56\ cm^2 = 5{,}56 \cdot 10^{-4}\ m^2$
$d = 5{,}56\ mm = 5{,}56 \cdot 10^{-3}\ m$
$K_1 = 7{,}00$
$K_2 = 12{,}0$
Dit : Berapa kapasitansinya?

Jawab :



$$C = C_1 + C_2$$

$$C = \frac{\varepsilon_0 \left(\frac{A}{2}\right) K_1}{d} + \frac{\varepsilon_0 \left(\frac{A}{2}\right) K_2}{d}$$

$$C = \frac{\varepsilon_0 \cdot A}{d} \left[\frac{K_1 + K_2}{2}\right]$$

Dipindai dengan CamScanner

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$$C = \frac{(8,85 \cdot 10^{-12})(5,56 \cdot 10^{-9})}{5,56 \cdot 10^{-3}} \left(\frac{7+12}{2}\right)$$

$$C = \frac{49,206 \cdot 10^{-16}}{5,56 \cdot 10^{-3}} \left(\frac{19}{2}\right)$$

$$C = 8,85 \cdot 10^{-13} (9,5)$$

$$C = 84,075 \cdot 10^{-13}$$

$$C = 8,91 \cdot 10^{-12} \text{ F} //$$

Dipindai dengan CamScanner